1-LABORATORIYA ISHI

Mavzu: NumPy asoslari va matritsalar bilan ishlash

Maqsad

Ushbu laboratoriya ishining maqsadi:

- NumPy kutubxonasi bilan ishlashni o‘rganish
- Vektor va matritsalar yaratish
- Statistik hisob-kitoblarni bajarish
- Real masala asosida ma’lumotlarni tahlil qilish
- Natijani CSV fayl ko‘rinishida saqlash

Kerakli vositalar

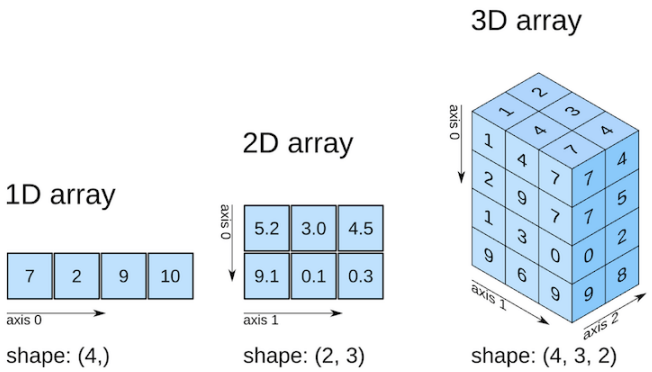
Talaba quyidagi vositalarga ega bo‘lishi kerak:

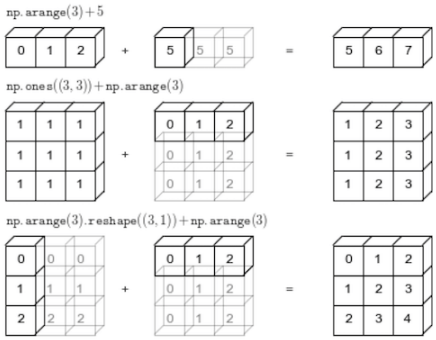
- Python 3.10 yoki undan yuqori
- NumPy kutubxonasi
- `pip install numpy`
- Jupyter Notebook / PyCharm / VS Code

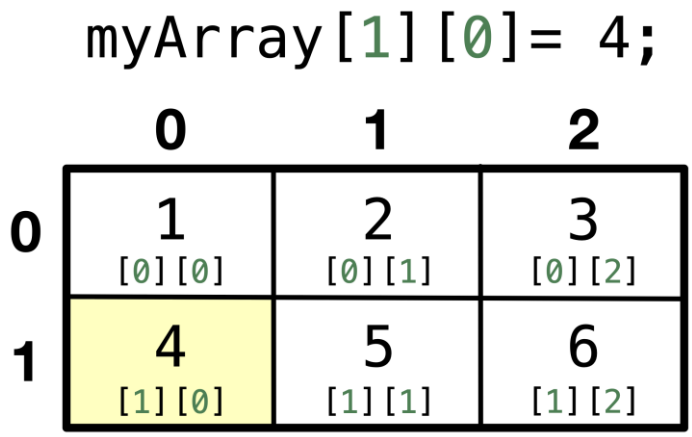
Nazariy asos

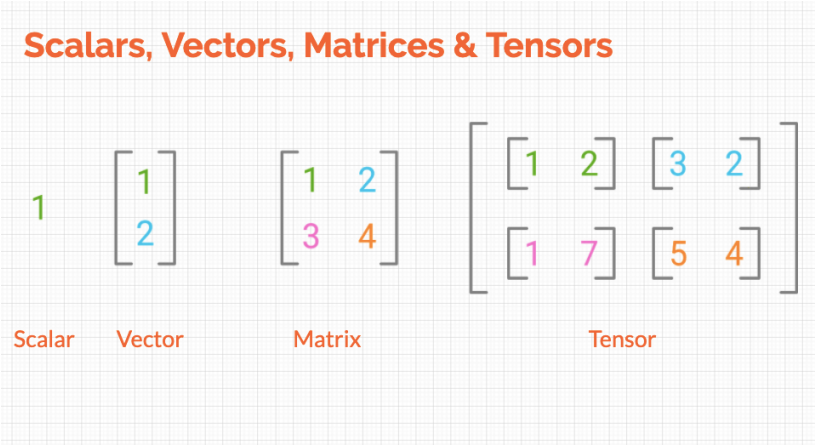
Sun’iy intellekt algoritmlari matritsa va vektorlar asosida ishlaydi. NumPy tezkor hisoblash uchun C asosida optimallashtirilgan kutubxona hisoblanadi.

NumPy array tuzilishi









Ishlash tartibi (Step-by-Step)

1-qadam: NumPy kutubxonasini ulash

```
import numpy as np
```

2-qadam: Real masala

Masala:

Universitetda 10 ta talabadan 5 ta fan bo‘yicha ball olingan.
Shu ma’lumotlar asosida:

1. O‘rtacha ballni hisoblang
2. Eng yuqori va eng past ballni aniqlang
3. Har bir talabani umumiy reytingini toping
4. Natijani CSV faylga saqlang

3-qadam: Ballar matritsasini yaratish

```
# 10 ta talaba, 5 ta fan
grades = np.random.randint(50, 100, (10, 5))

print("Talabalar ballari:")
print(grades)
```

4-qadam: Statistik hisob-kitob

```
print("Eng yuqori ball:", grades.max())
print("Eng past ball:", grades.min())
print("Umumiy o‘rtacha:", grades.mean())
```

5-qadam: Har bir talabani o‘rtacha reytingi

```
student_average = grades.mean(axis=1)

print("Har bir talabaning o‘rtacha bali:")
print(student_average)
```

6-qadam: Eng yaxshi talabani aniqlash

```
best_student = np.argmax(student_average)

print("Eng yaxshi talaba indeksi:", best_student)
```

7-qadam: Natijani CSV faylga saqlash

```
np.savetxt("students_grades.csv", grades, delimiter=",", fmt='%d')
```

To‘liq kod (Yakuniy variant)

```
import numpy as np

grades = np.random.randint(50, 100, (10, 5))

print("Talabalar ballari:\n", grades)

print("Eng yuqori ball:", grades.max())
print("Eng past ball:", grades.min())
print("Umumiy o‘rtacha:", grades.mean())

student_average = grades.mean(axis=1)
print("Talabalar o‘rtacha bali:", student_average)

best_student = np.argmax(student_average)
print("Eng yaxshi talaba:", best_student)

np.savetxt("students_grades.csv", grades, delimiter=",", fmt='%d')
```

Tekshirish bosqichi

Talaba quyidagilarni tekshirishi kerak:

✅ Matritsa 10x5 o‘lchamda yaratilgan
✅ O‘rtacha qiymatlar hisoblangan
✅ Eng yuqori va past ball chiqgan
✅ CSV fayl papkada paydo bo‘lgan
✅ Eng yaxshi talaba aniqlangan

Hisobot topshirish talabi

Talaba quyidagilarni topshiradi:

1. Python kodi (.py yoki .ipynb)
2. CSV fayl
3. Natijalar skrinshoti
4. 5-7 qatordan iborat tahlil (izoh)
    - Eng yuqori ball necha?
    - O‘rtacha ball qancha?
    - Qaysi talaba eng yaxshi?

Mezoni	Ball
Kod to‘g‘ri ishlashi	40
Hisob-kitob aniqligi	25
CSV saqlash	15
Tahlil yozish	10
Tartib va rasmiylik	10